

SHETH L.U.J AND SIR M.V COLLEGE

Aim:MANET Simulation with Directional Antenna

Simulate MANET using directional antennas and analyze improvements in spatial reuse and interference reduction.

Code:

```
# MANET Simulation using AODV (NS-2.35 Compatible)
```

```
set ns [new Simulator]
```

```
# Trace files
```

```
set tracefd [open manet.tr w]
```

```
$ns trace-all $tracefd
```

```
set namfd [open manet.nam w]
```

```
$ns namtrace-all-wireless $namfd 800 800
```

```
# Topography
```

```
set topo [new Topography]
```

```
$topo load_flatgrid 800 800
```

```
# Create GOD
```

```
create-god 5
```

```
# Wireless channel
```

```
set chan_1_ [new Channel/WirelessChannel]
```

```
# Node configuration
```

```
$ns node-config \
```

```
-adhocRouting AODV \
```

```
-llType LL \
```

```
-macType Mac/802_11 \
```

```
-ifqType Queue/DropTail/PriQueue \
```

```
-ifqLen 50 \
```

```
-antType Antenna/OmniAntenna \
```

```
-propType Propagation/TwoRayGround \
```

```
-phyType Phy/WirelessPhy \
```

```
-channel $chan_1_ \
```

```
-topoInstance $topo \
```

```
-agentTrace ON \
```

```
-routerTrace ON \
```

```
-macTrace ON \
```

```
-movementTrace ON
```

```
# Create nodes
```

```
for {set i 0} {$i < 5} {incr i} {
```

```
    set node_($i)[ns node]
```

```
}
```

```
# Initial positions
```

```
$node_(0) set X_ 100
```

```
$node_(0) set Y_ 100
```

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```
$node_(1) set X_ 250
$node_(1) set Y_ 200

$node_(2) set X_ 400
$node_(2) set Y_ 300

$node_(3) set X_ 550
$node_(3) set Y_ 400

$node_(4) set X_ 700
$node_(4) set Y_ 500

for {set i 0} {$i < 5} {incr i} {
    node_$i set Z_ 0
    $ns initial_node_pos node_$i 20
}

# UDP Agent
set udp0 [new Agent/UDP]
$ns attach-agent $node_(0) $udp0

# Null Agent
set null0 [new Agent/Null]
$ns attach-agent $node_(4) $null0

# Connect
$ns connect $udp0 $null0

# CBR Traffic
set cbr0 [new Application/Traffic/CBR]
$cbr0 set packetSize_ 512
$cbr0 set interval_ 0.1
$cbr0 attach-agent $udp0

# Mobility
nsat1.0"node_(1) setdest 300 300 10"
nsat2.0"node_(2) setdest 450 350 10"
nsat3.0"node_(3) setdest 600 450 10"

# Start traffic
nsat1.0"cbr0 start"
nsat8.0"cbr0 stop"

# Finish procedure
proc finish {} {
    global ns tracefd namfd

    $ns flush-trace

    close $tracefd
    close $namfd
    exec nam manet.nam &
```

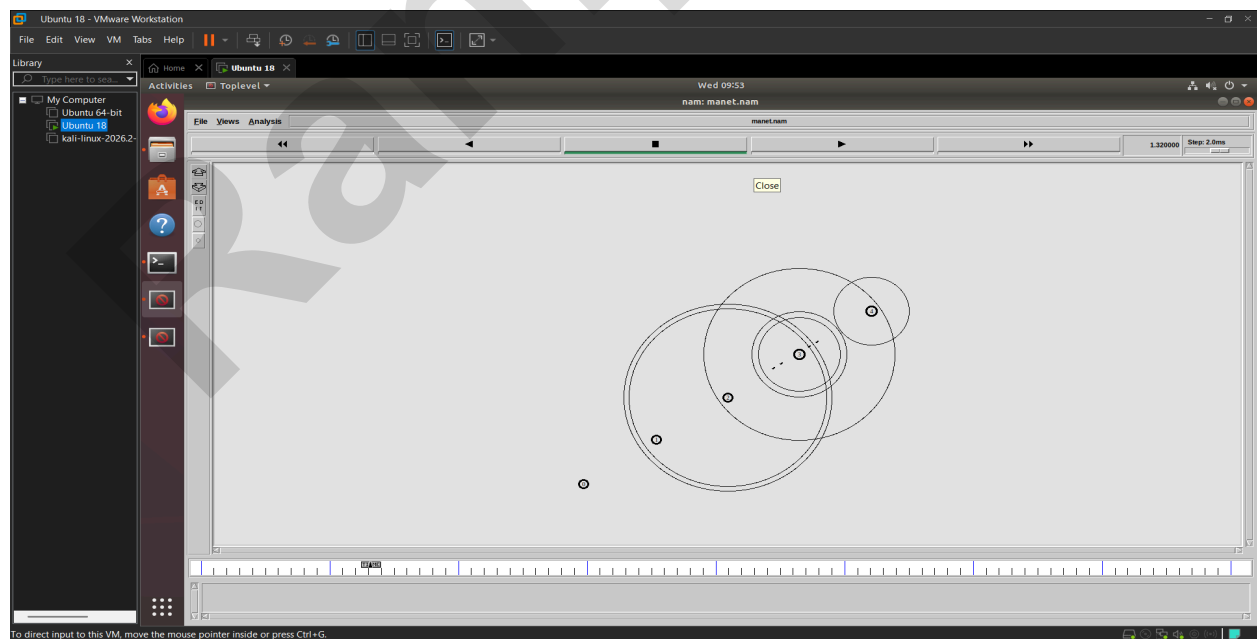
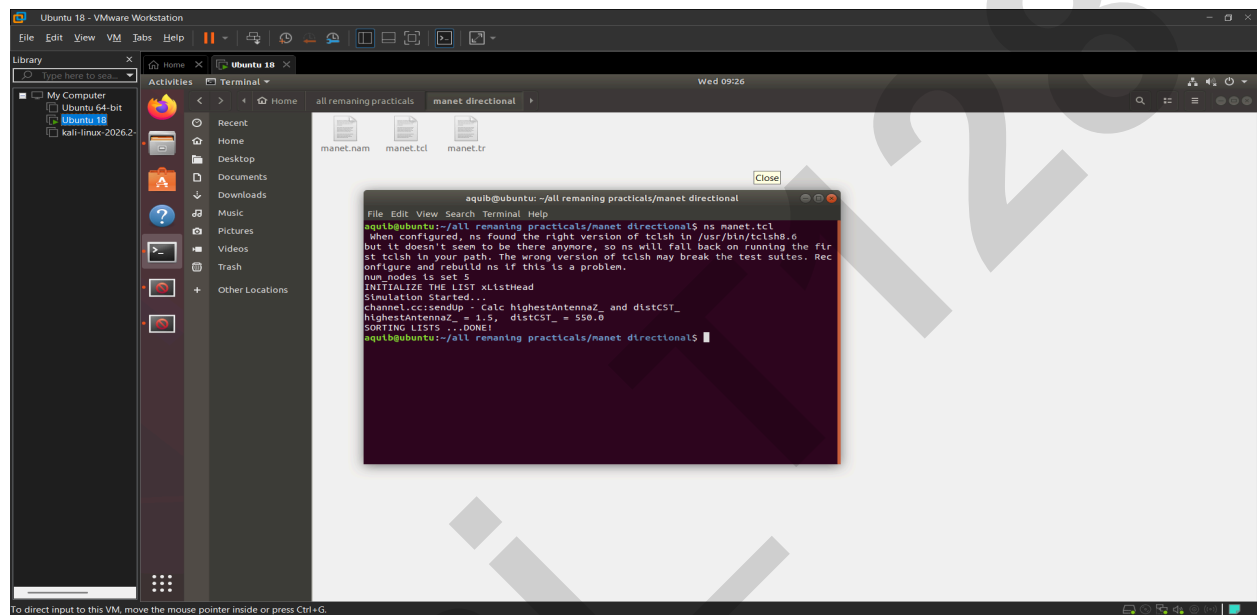
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```
    exit 0  
}
```

\$ns at 10.0 "finish"

puts "Simulation Started..."
\$ns run

OUTPUT:-



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